

ASSIGNMENT 3.2

VERSION CONTROL

1.0 Assignment Overview

In modern software development, programmers rarely work alone. Professional software projects require developers to collaborate, track changes, manage versions, and maintain project history using version control systems.

This assignment introduces students to collaborative development practices using Git and GitHub while continuing the semester-long calculator project.

Students will design and develop a modular calculator program using Python functions and manage the project using Git workflows such as commits, repositories, branching, push, pull, and collaboration practices.

This assignment combines:

- Python programming
- Modular design
- Version control
- Team collaboration
- Software documentation

2.0 Aim

To develop students' practical understanding of version control, collaborative programming workflows, and modular Python software development using Git and GitHub.

3.0 Learning Outcomes

By completing this assignment, students should be able to:

- Develop modular Python programs using functions.
- Apply version control techniques using Git.
- Create and manage repositories using GitHub.
- Track project changes using commits.
- Demonstrate collaborative programming workflows.
- Document software development processes professionally.

4.0 Scenario-Based Case Study

Scenario

You have recently joined a software development company called **SmartCalc Technologies** as a junior developer.

The company is developing a command-line calculator application for students learning Python programming.

COMPUTING PROGRAMMING (COM4006)

Your team has been instructed to:

- Develop the calculator using modular programming principles
- Track all project changes using Git
- Store the project on GitHub
- Collaborate using professional software development workflows

The project manager has explained that all developers must use:

- Repositories
- Commits
- Branching
- Push and pull operations
- Collaboration workflows

Your task is to design, develop, manage, and document the calculator project professionally.

5.0 Assignment Tasks

Part A – Repository Setup and Configuration

Task Requirements

1. Install Git on your computer.
2. Configure Git using:
 - Username
 - Email address
3. Create a local Git repository named:

smartcalc_project

4. Create a GitHub repository for the project.
5. Connect the local repository to GitHub.

Evidence Required

Provide screenshots showing:

- Git installation verification
- Git configuration

COMPUTING PROGRAMMING (COM4006)

- Local repository creation
- GitHub repository creation

Part B – Modular Calculator Development**Task Requirements**

Develop a modular Python calculator program using separate functions.

Your calculator must include:

- Addition
- Subtraction
- Multiplication
- Division

Each operation must:

- Use a separate function
- Accept parameters
- Return a result

Additional Requirements

Your program must:

- Display a menu
- Accept user input
- Handle invalid choices
- Prevent division by zero
- Use meaningful variable names
- Include comments

Extension Features

Add at least TWO additional features such as:

- Power calculation
- Percentage calculation
- Square root
- Average calculation
- Continuous calculator loop

COMPUTING PROGRAMMING (COM4006)**Part C – Git Commit Workflow****Task Requirements**

You must demonstrate proper Git usage throughout development.

Create commits for:

- Initial project setup
- Addition function
- Subtraction function
- Multiplication function
- Division function
- Extension features
- Final version

Commit Message Requirements

Use meaningful commit messages.

Good Examples

- Added addition function
- Fixed division by zero error
- Added calculator menu

Poor Examples

- Update
- Stuff
- Changes

Part D – Collaboration and Branching**Task Requirements**

Simulate collaborative development by using branches.

1. Create a branch named:

new-feature

2. Add a new calculator feature inside the branch.
3. Commit the changes.
4. Merge the branch into the main branch.

COMPUTING PROGRAMMING (COM4006)

5. Push the updated project to GitHub.

Evidence Required

Provide screenshots showing:

- Branch creation
- Branch switching
- Merge operation
- Updated GitHub repository

Part E – Reflective Report

Write a short reflective report (500–700 words) discussing:

- What is version control?
- Why are commits important?
- What is the difference between Git and GitHub?
- How does Git support teamwork?
- Challenges faced during the assignment
- Skills learned during the project

6.0 Submission Requirements

Students must submit:

1. Python source code (.py)
2. GitHub repository link
3. Screenshots of:
 - Git commands
 - Commits
 - Branches
 - GitHub repository
4. Reflective report (PDF or Word)

COMPUTING PROGRAMMING (COM4006)

7.0 Academic Integrity

Students must:

- Write their own code
- Acknowledge external sources where necessary
- Avoid copying from classmates or online repositories

8.0 Recommended Resources

1. Matthes, E. (2019) *Python Crash Course*
2. Sweigart, A. (2020) *Automate the Boring Stuff with Python*
3. Chacon, S. and Straub, B. (2014) *Pro Git*
4. [Git Documentation](#)
5. [GitHub Documentation](#)

9.0 Conclusion

This assignment introduces students to professional software development practices using modular programming, Git version control, and GitHub collaboration workflows. By completing the assignment, students will gain practical experience in managing and developing software projects using industry-standard tools and techniques.